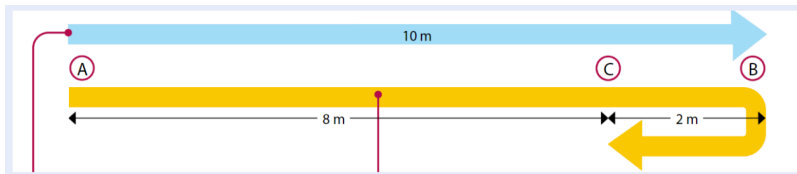


Kinematics Main Note

Speed

Differences between Distance and Displacement



Distance	Displacement
Total length covered by moving object, regardless of direction of motion ⇒ has magnitude only (no direction) → so it is SCALAR.	Distance measured in a straight line from a fixed reference point. ⇒ has both magnitude and direction. → so it is VECTOR.
SI unit: meter (m)	SI unit: meter (m) ⇒ refers to shortest distance and direction of object from starting point. → can be negative. (object moves in negative direction from starting point)

When is displacement negative:

1. moving in opposite direction compared to initial position
2. Returning to initial position from positive displacement
3. motion in negative direction from initial point

Velocity

Acceleration

Displacement time graphs

Displacement time graphs

Velocity Time graphs

Velocity time graphs

Important:

⇒ Gravity is the force that pulls both objects down ⇒ both object with lower and higher mass will have same initial acceleration.

Terminal velocity ⇒ depends on weight of object (terminal velocity is reached when the weight is balanced with air resistance) ⇒ greater air resistance is required to balance greater weight and thus > heavier object has higher terminal velocity

- as AR increases with increased velocity ⇒ which causes acceleration to decrease.

Example Questions

1. Car is travelling 10m/s due north. it reaches a bend and short time later it travels at 10m/s due east. explain why car changes velocity and accelerates.

- speed (numerical value) remains same, but velocity changes as its direction changes. since there is a change in velocity, there must be acceleration
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